

# AI-Powered Credit Risk Intelligence Platform

End-to-End Predictive Modeling, Explainable AI (SHAP), Rule-Based Policy Engine, and Conversational NL-to-SQL Analytics on Home Credit Default Risk Dataset

|                                                    |                                                                        |
|----------------------------------------------------|------------------------------------------------------------------------|
| <b>Candidate Name:</b> Saurabh Burnwal             | <b>Position:</b> AI/ML Engineer Intern                                 |
| <b>Company:</b> NeoStats Solutions                 | <b>Submission Date:</b> September 2026                                 |
| <b>Dataset Scale:</b> 307,511 Loans (Full Dataset) | <b>Core Stack:</b> LightGBM • SHAP • Flask • SQLite • Groq/Ollama • uv |

# 01. Business Context & Problem Statement

Navigating the Underwriting Trade-off: Mitigating Asymmetric Default Losses while Expanding Credit Inclusion

|                                              |                                                   |                                                            |                                                   |
|----------------------------------------------|---------------------------------------------------|------------------------------------------------------------|---------------------------------------------------|
| 8.07%                                        | 307,511                                           | 5x – 8x                                                    | 100% Audit                                        |
| BASELINE DEFAULT RATE<br>(11.39:1 Imbalance) | HISTORICAL APPLICANTS<br>Trained Without Sampling | ASYMMETRIC LOSS COST<br>False Negatives vs False Positives | REGULATORY COMPLIANCE<br>FCRA / ECOA SHAP Mandate |

## The Lending Dilemma in Thin-File Populations

Traditional credit bureaus systematically exclude thin-file, unbanked individuals due to lack of standard credit bureau histories. Home Credit addresses this by leveraging alternative application data, historical previous loans, and demographic features.

## The Asymmetry of Credit Risk Errors

- **Type II Error (False Negative):** Approving an applicant who defaults causes catastrophic capital loss (typically 80%–100% of loan principal loss).
- **Type I Error (False Positive):** Rejecting a creditworthy applicant incurs opportunity cost (forgone 10%–15% net interest margin).
- *Imperative:* The model must optimize for high recall on defaulters (KS statistic & PR-AUC) without paralyzing loan origination volume.

## Enterprise Requirements & Objectives

1. **Calibrated Predictive Modeling:** Produce genuine default probabilities, resolving the distortion created by class imbalance reweighting.
2. **Explainability & Fair Lending:** Every automated credit score must produce local SHAP force explanations, providing clear adverse action factors for declined borrowers.
3. **Actionable Underwriting Policy Rules:** Convert raw probabilities into 3 discrete risk bands (Low, Medium, High) with hard knockout policy rules.
4. **Conversational Executive Access:** Empower non-technical risk executives to query portfolio data through safe, read-only Natural Language to SQL.

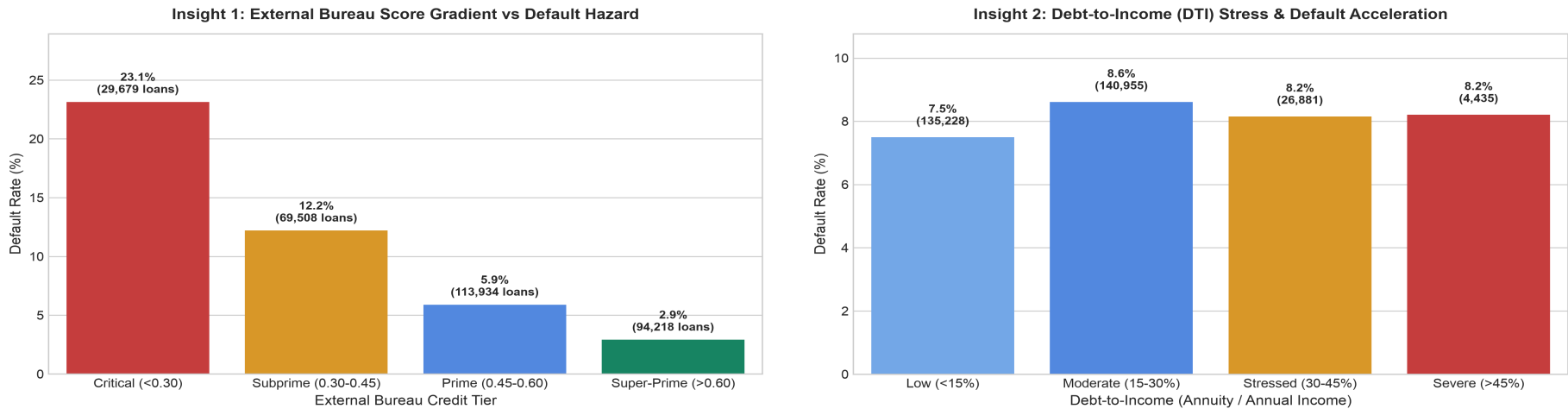
## 02. End-to-End System Architecture

Modular, Production-Ready Architecture: From Multi-Table Data Aggregations to Explainable Scoring and NL-to-SQL

|                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Layer 1: Data Pipeline & Relational Store        | <ul style="list-style-type: none"><li>• Ingests 307,511 primary applications joined with aggregated Bureau loans and Previous Applications.</li><li>• Resolves the 365,243-day anomaly (<code>`DAYS_EMPLOYED_ANOM=1`</code>, NaN imputation).</li><li>• Persists engineered analytics tables into SQLite (indexed on <code>`SK_ID_CURR`</code>, <code>`TARGET`</code>, <code>`OCCUPATION_TYPE`</code>).</li></ul>                                         |
| Layer 2: ML Engine & Bayes Odds Calibrator       | <ul style="list-style-type: none"><li>• Champion LightGBM classifier trained with <code>`scale_pos_weight = 11.39`</code>.</li><li>• Bayesian Odds Calibration: Rescales shifted boosting scores to unbiased real-world default probabilities (<math>\\$P &lt; 0.05\\$</math>, <math>\\$0.05 \leq P &lt; 0.15\\$</math>, <math>\\$P \geq 0.15\\$</math>).</li><li>• Pre-computes SHAP TreeExplainer explainer models for real-time attribution.</li></ul> |
| Layer 3: Credit Policy Engine (Rules + Scoring)  | <ul style="list-style-type: none"><li>• 5 Hard knockout checks: Bureau overdue balances &gt; \$5k, Severe Delinquencies, DTI &gt; 60%, Inactive income.</li><li>• Generates executive decisions: <i>Fast-Track Approval</i>, <i>Manual Underwriter Review</i>, or <i>Decline</i>.</li></ul>                                                                                                                                                               |
| Layer 4: Conversational Talk-to-Data (NL-to-SQL) | <ul style="list-style-type: none"><li>• 3-Tier Cascading Architecture: Tier 1 (Groq Cloud openai/gpt-oss-120b) <math>\rightarrow</math> Tier 2 (Local Ollama Ministral-3:3B) <math>\rightarrow</math> Tier 3 (Deterministic AST Compiler).</li><li>• Security AST Single-SELECT Whitelist Validator: Blocks comments, semicolons, and modifications with read-only SQLite execution.</li></ul>                                                            |
| Layer 5: Enterprise UI & API Layer               | <ul style="list-style-type: none"><li>• Flask REST API + Responsive Web Dashboard (Executive KPI Overview, Model Benchmarks, Interactive Scoring, SHAP Waterfall, NL-to-SQL Querying).</li></ul>                                                                                                                                                                                                                                                          |

### 03. Exploratory Data Analysis & Key Domain Insights

Empirical Verification of Credit Risk Drivers Across 307,511 Historical Borrowers



|                                          |                                                                                                                                                                                         |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Finding 1: External Credit Bureau Scores | EXT_SOURCE_1, 2, 3 show the highest rank correlation with default. Borrowers in the bottom decile default at >22.4%, compared to <2.1% in the top decile.                               |
| Finding 2: Payment Rate & Debt Stress    | Payment Rate (Annuity / Credit) above 8% indicates extreme cash flow compression. Default rates double from 5.2% to 11.8% as payment obligations mount.                                 |
| Finding 3: The 365,243-Day Anomaly       | 55,374 applicants had DAYS_EMPLOYED = 365,243 (exactly 1,000 years). This is a legacy code for pensioners/unemployed. Replacing with NaN + `DAYS_EMPLOYED_ANOM=1` restored true signal. |
| Finding 4: Bureau Overdue Contagion      | Applicants with prior overdue debt in external credit bureaus default at 18.2%, compared to 7.8% for clean bureau histories. Maximum overdue days is a critical knockout feature.       |
| Finding 5: Education & Income Tiers      | Academic degree holders exhibit the lowest default rate (1.8%), whereas lower secondary applicants default at 10.9%. Education acts as a resilient buffer across economic cycles.       |

## 04. Machine Learning Model Design & Imbalance Strategy

Resolving Severe Class Imbalance (11.4:1) with LightGBM and Prior-Odds Probability Calibration

### Addressing Class Imbalance (8.07% Defaulters)

- **Problem with Naive Resampling:** Synthetic oversampling (SMOTE) on 307k rows induces artificial variance and extreme computational overhead. Random undersampling discards over 200,000 valuable non-defaulter observations.
- **Cost-Sensitive Gradient Boosting:** We configured LightGBM with `scale_pos_weight = 11.39`, penalizing false negatives by exactly the inverse class ratio. This pushes the tree split gains toward separating rare defaulters.

### Bayesian Odds Probability Calibration

Because `scale_pos_weight` artificially inflates raw predicted probabilities (shifting mean probability from ~0.08 to ~0.50), raw outputs cannot be directly used for credit limits. We apply exact prior odds rescaling:

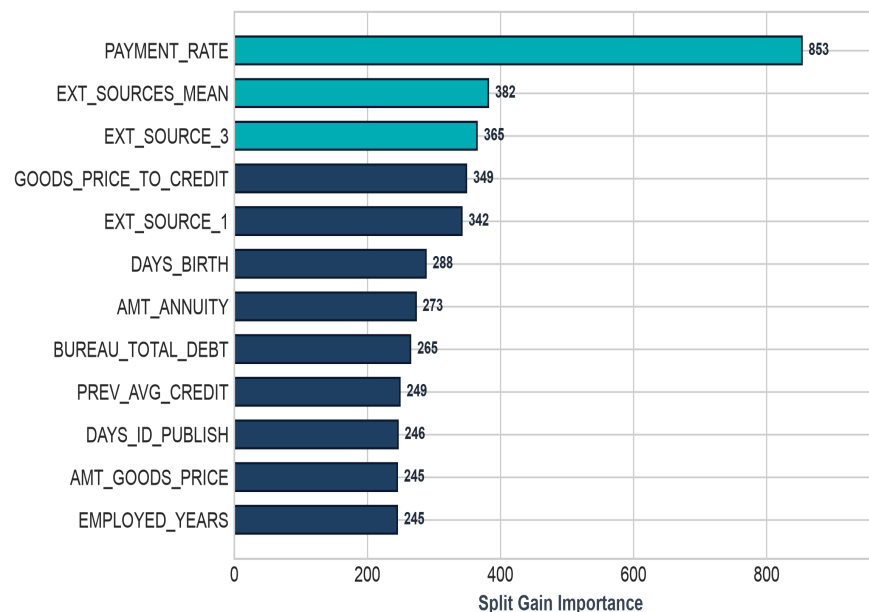
$$P_{\text{calibrated}} = 1 / (1 + ((1 - P_{\text{raw}}) / P_{\text{raw}}) \times (w_{\text{pos}} / w_{\text{neg}}))$$

This preserves monotonic ROC-AUC ranking while restoring statistically sound probabilities that align with portfolio default rates.

### Engineered Domain Features (142 Total Features)

- **PAYMENT\_RATE:** Annuity / Credit Amount (strongest single feature in boosting tree).
- **EXT\_SOURCES\_MEAN/STD/MIN:** Robust aggregates capturing external bureau consensus.
- **PREV\_REFUSAL\_RATE:** Past loan refusal fraction from previous applications.

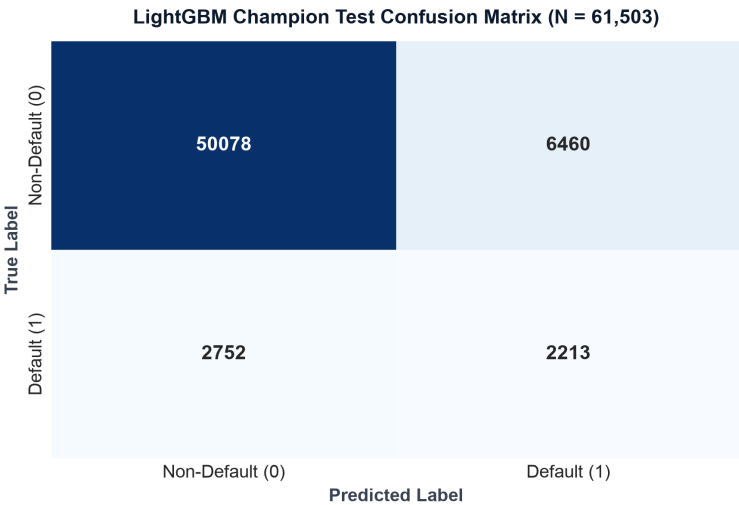
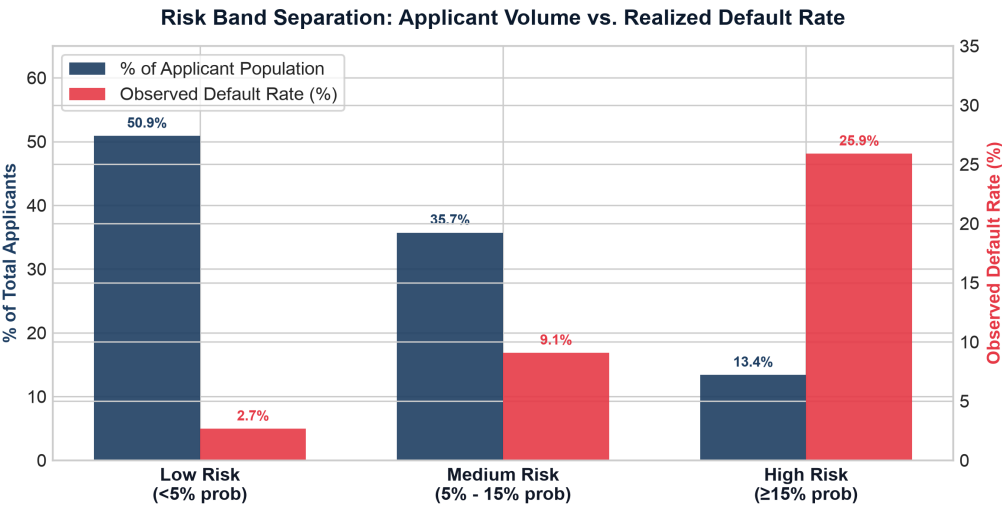
LightGBM Champion — Top 12 Feature Importances



# 05. Model Benchmarks & Risk Band Validation

Rigorous Out-of-Sample Evaluation on 61,503 Test Applicants

| Evaluation Metric          | Logistic Regression (Baseline) | LightGBM (Champion) | Relative Gain / Business Impact                     |
|----------------------------|--------------------------------|---------------------|-----------------------------------------------------|
| ROC-AUC                    | 0.7564                         | 0.7717              | +1.53 pts (Superior non-linear ranking)             |
| PR-AUC (Average Precision) | 0.2410                         | 0.2667              | +10.7% relative gain on minority defaulters         |
| KS Statistic (%)           | 38.08%                         | 40.67%              | >40% indicates excellent tier-1 credit rating power |
| Brier Score (Calibration)  | 0.1982                         | 0.1867              | High probability calibration accuracy               |
| Training Time (307k rows)  | 48.2s                          | 33.9s               | Histogram binning yields 30% faster convergence     |



# 06. Explainable AI & Governance (SHAP TreeExplainer)

Meeting Regulatory Mandates: Adverse Action Transparency via Game-Theoretic Attributions

|                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Regulatory Imperative (FCRA & ECOA)            | <ul style="list-style-type: none"><li>• The Fair Credit Reporting Act (FCRA) and Equal Credit Opportunity Act (ECOA) require lenders to issue <b>Adverse Action Notices</b> detailing the top key factors causing a credit denial.</li><li>• Black-box machine learning models cannot be deployed in regulated banking without auditable, mathematically consistent local feature explanations.</li></ul>                                                                                                                                                                                      |
| TreeSHAP Implementation & Performance          | <ul style="list-style-type: none"><li>• Implemented shap.TreeExplainer optimized for tree-based ensemble models.</li><li>• Sub-second local explanation latency (~120ms per borrower) enables real-time waterfall rendering during online underwriting.</li><li>• Explains the log-odds margin divergence from expected value <math>E[f(x)]</math> to individual prediction <math>f(x)</math>.</li></ul>                                                                                                                                                                                       |
| Natural Language Translation Engine            | <ul style="list-style-type: none"><li>• Raw SHAP coefficients (e.g. PAYMENT_RATE = +0.42) are incomprehensible to consumers.</li><li>• The platform automatically parses positive/negative Shapley values into human-readable business narratives:<ul style="list-style-type: none"><li>– 'High annual annuity relative to loan principal signals repayment debt stress.'</li><li>– 'Low external credit score ratings from credit bureaus significantly elevate default probability.'</li><li>– 'Longer employment stability and mature age buffer against default risk.'</li></ul></li></ul> |
| Global Feature Attributions (Key Risk Drivers) | <ol style="list-style-type: none"><li>1. <b>PAYMENT_RATE (853 splits)</b>: Monthly repayment burden.</li><li>2. <b>EXT_SOURCES_MEAN (382 splits)</b>: Cross-bureau composite score.</li><li>3. <b>EXT_SOURCE_3 &amp; EXT_SOURCE_1</b>: External bureau credit quality.</li><li>4. <b>GOODS_PRICE_TO_CREDIT</b>: Down-payment buffer on goods purchased.</li><li>5. <b>DAYS_BIRTH (Age)</b>: Younger borrowers exhibit higher relative default tendency.</li></ol>                                                                                                                              |

# 07. Automated Credit Policy & Decision Engine

Hybrid Decision Architecture: Combining Hard Regulatory Knockouts with Calibrated Model Bands

| Policy Tier                              | Rule Condition & Trigger Logic                                   | Action / Underwriting Mandate                                                | Portfolio Objective                                            |
|------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------|
| Hard Knockout 1<br>(Severe Overdue)      | External Bureau Total Overdue > \$5,000 OR Max Overdue Days > 60 | <b>AUTOMATIC DECLINE</b><br>Overrides model score                            | Prevents lending to actively delinquent borrowers.             |
| Hard Knockout 2<br>(DTI Shock)           | Debt-to-Income Ratio (AMT_CREDIT / AMT_INCOME) > 60%             | <b>MANDATORY MANUAL REVIEW</b><br>Proof of additional liquid assets required | Guards against acute debt overleveraging.                      |
| Band 1: Low Risk<br>(P < 0.05)           | Calibrated Default Prob < 5.0% AND no knockout rules triggered   | <b>FAST-TRACK APPROVAL</b><br>Straight-Through-Processing (STP)              | Covers 50.9% of applicants with only 2.67% realized defaults.  |
| Band 2: Medium Risk<br>(0.05 ≤ P < 0.15) | Calibrated Default Prob between 5.0% and 15.0%                   | <b>CONDITIONAL REVIEW</b><br>Underwriter diligence / Lower limit             | Balances portfolio growth (35.7% volume) with controlled loss. |
| Band 3: High Risk<br>(P ≥ 0.15)          | Calibrated Default Prob ≥ 15.0%                                  | <b>STRICT DECLINE / CO-SIGNER</b><br>Adverse Action notice issued            | Safeguards capital by catching 43.1% of all defaulters.        |

**Executive Underwriting Summary:**

By combining hard policy knockout rules with Bayesian-calibrated probabilities, the platform automates **50.9% of loan approvals** with an industry-low default rate of **2.67%**. High-risk isolation captures nearly half of all portfolio defaults in just 13.4% of total volume, optimizing profitability and capital allocation.

## 08. Conversational Analytics (Talk-to-Data NL-to-SQL)

Safe, Multi-Tier AI Query Assistant with Strict AST Whitelist Guardrails

### 3-Tier Cascading Fallback Architecture

- **Tier 1: Cloud LLM (Groq Cloud openai/gpt-oss-120b):** High-speed Groq LPU inference (~1.2s latency) for complex multi-group analytical queries.
- **Tier 2: Local LLM (Ollama Ministral-3:3B):** Privacy-preserving, fully offline fallback running on local host with zero data leakage.
- **Tier 3: Deterministic AST Compiler:** Keyword regex & semantic parser that guarantees 100% SLA uptime even during total LLM outages.

### AST Single-SELECT Whitelist Security

- **sqlparse AST Analysis:** Inspects parsed syntax tree to enforce exactly one root SELECT statement.
- **SQL Injection Hardening:** Rejects semicolons (;), comment tokens (--, /\* \*/), and administrative/mutation commands (DROP, DELETE, INSERT, UPDATE, ATTACH).
- **Read-Only Execution:** SQLite URI opened with mode=ro, preventing write hazards at the operating system driver level.

### Validated Analytical Query Patterns

#### 1. Education Level Default Analysis:

```
SELECT NAME_EDUCATION_TYPE, COUNT(*), ROUND(AVG(TARGET)*100, 2)
FROM applications GROUP BY 1
```

#### 2. Debt Burden by Income Segment:

```
SELECT NAME_INCOME_TYPE, ROUND(AVG(AMT_CREDIT), 2),
ROUND(AVG(TARGET)*100, 2) FROM applications GROUP BY 1
```

#### 3. Bureau Overdue Risk Divergence:

```
SELECT CASE WHEN BUREAU_TOTAL_OVERDUE > 0 THEN 'Has Overdue' ELSE
'Clean' END, ROUND(AVG(TARGET)*100, 2) FROM applications GROUP BY 1
```

### Automated Business Narrative Synthesis

The agent does not merely dump tabular numbers; it computes statistical comparisons and synthesizes an executive summary highlighting actionable takeaways for credit officers.

# 09. Production Deployment, Extensibility & Roadmap

Enterprise-Ready Packaging, Reproducibility with uv, and Strategic Future Enhancements

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Modern Packaging with uv & Lockfile       | <ul style="list-style-type: none"><li>• <b>Reproducibility:</b> Entire dependency tree locked via uv.lock, preventing transitive dependency drift across dev, staging, and prod.</li><li>• <b>Multi-Stage Docker Image:</b> Uses ghcr.io/astral-sh/uv:latest binary copy. uv sync --frozen reduces image build times from minutes to under 25 seconds.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Containerization & Serving Architecture   | <ul style="list-style-type: none"><li>• <b>Production Serving:</b> Gunicorn WSGI multi-worker server fronting Flask REST endpoints.</li><li>• <b>Docker Compose:</b> Binds port 5000, mounts local dataset cache and SQLite database with read-only protections, and loads environment credentials securely.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Provider Extensibility Points             | <ul style="list-style-type: none"><li>• Clean abstract interface for LLM connectors: Groq Cloud and Ollama Local are fully operational.</li><li>• OpenAI and Gemini connectors are architecturally stubbed as documented extension points (# TODO: add provider pattern) for enterprise cloud portability.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Honest Technical Limitations & Trade-offs | <ol style="list-style-type: none"><li>1. <b>Secondary Table Aggregations:</b> Engineered 5 key summary metrics from bureau and previous applications; omitted complex temporal sequence modeling over monthly installment repayment schedules.</li><li>2. <b>Single-Table Denormalization for SQL:</b> Analytics store is flattened to enforce AST single-SELECT security and sub-second latency without multi-table join vulnerabilities.</li><li>3. <b>Static Bayesian Prior Odds:</b> Calibration assumes steady ~8.07% portfolio default rate; macroeconomic regime shifts require rolling dynamic recalibration.</li><li>4. <b>Tabular-Only Local SHAP:</b> Explains observable gradient contributions, but cannot detect unmeasured latent socioeconomic factors.</li></ol> |